OLS Inference on Synthetic Data with Generative AI

When it is valid, and how to make it valid when it is not

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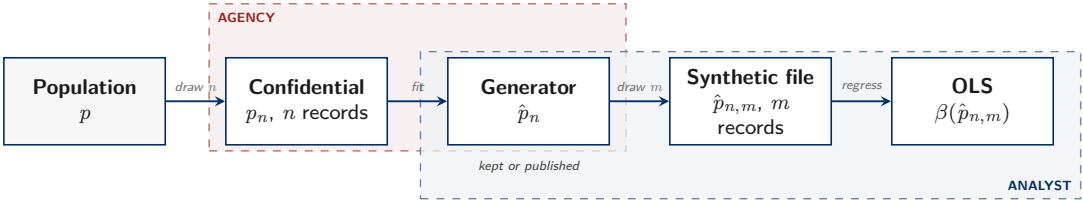
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Ph.D. Thesis Defense



Why Synthetic Data?

- Tax records, patient files and firm surveys underlie much applied work, but access to them is tightly restricted.
- Deep generative models now produce realistic **synthetic records**, and agencies release these instead.
- Analysts then run OLS on them as if they were real. But the records are draws from a *fitted model*, so they carry the model's error on top of sampling noise.
- Releases come with fit diagnostics, not with guarantees for standard errors or confidence intervals.



Question. When is OLS inference on synthetic data valid, and how do we make it valid when it is not?

Related Literature

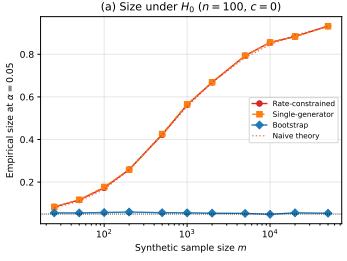
Strand	Key references	What is missing
Multiple imputation	Rubin (1993), Raghunathan et al. (2003)	Assumes a posterior-predictive generator
Flow-based generators	Liu et al. (2023), Sahoo et al. (2026)	Rates, but no inference
Bootstrap for regression	Bickel and Freedman (1981), Freedman (1981)	No generator bias
Privacy auditing	Shokri et al. (2017), Platzer and Reutterer (2021)	Leakage, not validity

A Gaussian Example: A Generator Without Bias

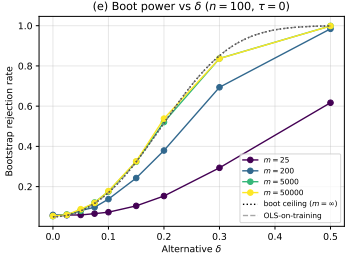
Data $Y_i \sim \mathcal{N}(\mu, 1)$; generator $\hat{p}_n = \mathcal{N}(\hat{\mu}_n, \hat{\sigma}_n^2)$; estimator $\hat{\theta}_m$ = mean of the m synthetic draws.

$$\mathrm{Var}(\hat{\theta}_m) \;=\; \underbrace{\tfrac{1}{n}}_{\substack{\text{training sample} \\ \text{(new data, new } \hat{\mu}_n\text{)}}} \;+\; \underbrace{\tfrac{1}{m}}_{\substack{\text{Monte Carlo} \\ \text{(new draws)}}}$$

Treating the synthetic file as real data sees only $\frac{1}{m}$. Retraining the generator also captures $\frac{1}{n}$.



Plug-in s.e.: size goes to 1 as m grows. Bootstrap: stays at 5%.



Bootstrap power reaches that of OLS on the confidential data.

What OLS Requires from the Generator

For any distribution q of (X, Y), let $\beta(q) := \Sigma_{XX}(q)^{-1}\Sigma_{XY}(q)$, with $\Sigma_{XX}(q) = \mathbb{E}_q[XX']$ and $\Sigma_{XY}(q) = \mathbb{E}_q[XY]$, be the least-squares coefficient under q. Population: $\beta = \beta(p)$; confidential OLS: $\beta(p_n)$; synthetic OLS: $\beta(\hat{p}_{n,m})$. The score is $\mathrm{sc}(q) := \mathbb{E}_q[X(Y - X'\beta)]$.

$$\beta(\hat{p}_{n,m}) - \beta \;=\; \underbrace{\big(\beta(\hat{p}_{n,m}) - \beta(\hat{p}_n)\big)}_{\text{(A) Monte Carlo}} \;+\; \underbrace{\Sigma_{XX}(\hat{p}_n)^{-1}\,\big[\,G_n + \mathrm{sc}(p_n)\,\big]}_{\text{(B) generator error}}$$

$G_n = \Delta_{XY}(p_n) - \Delta_{XX}(p_n)\beta$ is the gap between the generator and the training sample, with $\Delta_{XX}(p_n) := \Sigma_{XX}(\hat{p}_n) - \Sigma_{XX}(p_n)$ and $\Delta_{XY}(p_n) := \Sigma_{XY}(\hat{p}_n) - \Sigma_{XY}(p_n)$.

Theorem (Oracle condition). If $m/n \to \infty$ and $\|\Delta_{XX}(p_n)\|, \|\Delta_{XY}(p_n)\| = o_p(n^{-1/2})$, then synthetic OLS is equivalent to OLS on the confidential data:

$$\sqrt{n}\,\big(\beta(\hat{p}_{n,m}) - \beta(p_n)\big) \xrightarrow{p} 0.$$

The Gaussian example is the case $G_n \equiv 0$.

Flow Generators Sit at the Boundary

The W_2 distance to the population bounds the moment errors, $\|\Delta_{XX}(p)\|, \|\Delta_{XY}(p)\| = O_p\big(W_2(\hat{p}_n, p)\big)$.

Generator	$W_2(\hat{p}_n, p)$	Consequence
Flow Matching	$\widetilde{O}_p(n^{-1/4})$	bias dominates, $\sqrt{n}$ rate lost
Rectified Flow	$\widetilde{O}_p(n^{-1/2})$	generator error $\asymp$ sampling noise

Benton et al. (2024); Gaur et al. (2025); Sahoo et al. (2026)

Theorem (Rate-constrained inference). If $W_2(\hat{p}_n, p) = O_p(n^{-\nu})$ and $m = o(n^{2\nu})$, then $\sqrt{m}\,(\beta(\hat{p}_{n,m}) - \beta) \xrightarrow{d} \mathcal{N}(0, V)$. Valid, but it throws away synthetic data and the right m cannot be chosen from the data.

Inference at the Boundary: Bootstrap That Retrains the Generator

Assumption 4 (Generator score CLT).

$$\sqrt{n}\begin{pmatrix}\mathrm{sc}(\hat{p}_n)\\ \mathrm{sc}(p_n)\end{pmatrix} \xrightarrow{d} \mathcal{N}\left(0, \begin{pmatrix}\Omega_{\text{total}} & \Omega'_{\text{cross}}\\ \Omega_{\text{cross}} & \Omega\end{pmatrix}\right), \qquad \sup_n \mathbb{E}\left\|\sqrt{n}\,\mathrm{sc}(\hat{p}_n)\right\|^{1+\delta} < \infty.$$

The Rectified Flow rate only gives $\sqrt{n}\,\|\,\mathrm{sc}(\hat{p}_n)\| = O_p(\log^k n)$; Assumption 4 gives this bound a centred Gaussian limit.

Theorem (Total error). If $m/n \to \infty$, then $\sqrt{n}\,(\beta(\hat{p}_{n,m}) - \beta) \xrightarrow{d} \mathcal{N}(0, V_{\text{total}})$ with $V_{\text{total}} = V + V_{\text{gen}} + C_{\text{cross}}$.

Bootstrap, $b = 1, \ldots, B$

Resample n records, **retrain** the generator, draw $m^* \gg n$ and compute β^{*b}. Then $n\,\widehat{\mathrm{Var}}(\beta^{*b}) \xrightarrow{p} V_{\text{total}}$.

Simulations: Only the Bootstrap Attains Nominal Coverage

Single regressor ($K = 1$) and 15 correlated regressors ($K = 15$, joint Wald test). $n = 2{,}500$, $m = 20{,}000$.

Variance strategy	What changes across replications	$K = 1$	$K = 15$
Nonparametric bootstrap	Resample data, retrain generator	**0.94**	**0.93**
Parametric bootstrap	Draw from $\hat{p}_n$, retrain generator	0.89	0.94
Multi-generator	Same data, new seeds	0.80	0.09
Multi-dataset	One generator, new draws	0.34	0.00

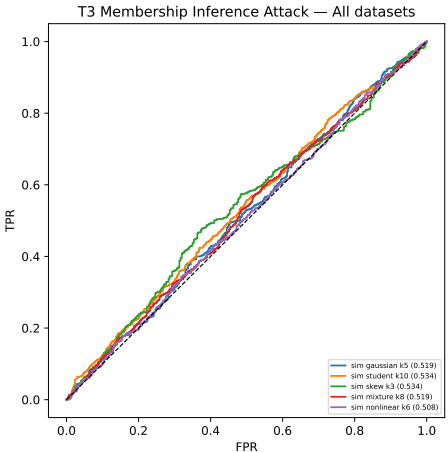
Coverage of nominal 95% intervals, Rectified Flow.

Privacy: Learning the Population, Not the Records

Can an attacker tell whether a record was used to train the generator (Platzer and Reutterer, 2021)?

Membership-inference AUC: **0.51–0.53**. A coin flip gives 0.5.

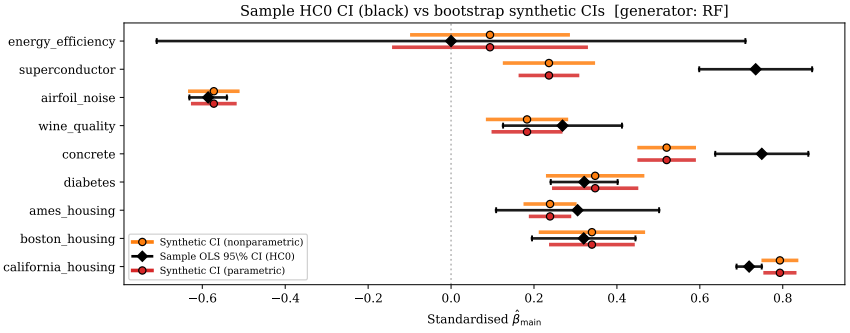
> **Privacy through generalisation**
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> Privacy is guaranteed by the generalisation of the learned distribution instead of data replacement.



ROC curves lie on the diagonal

Applications: Nine Real Datasets

Housing, medicine, engineering and physics data, $n = 442$ to $21{,}263$. Black: OLS on the real data. Orange/red: synthetic 95% intervals.



Sign and significance of the main coefficient are recovered in **8 of 9** datasets.

Conclusion

- Efficient synthetic OLS needs the generator to match the training sample's second moments to $o_p(n^{-1/2})$.
- Rectified Flow only reaches this rate at the boundary, so plug-in standard errors are too small.
- A bootstrap that retrains the generator restores nominal coverage, and the generator does not memorise records.

For practice

Release several generators retrained on resamples, not a single synthetic file.

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